PHASE 1: BUSINESS AND DATA UNDERSTANDING

- ## Business Understanding Notes

Business Understanding

Road traffic accidents are one of the leading causes of injuries, fatalities, and economic losses worldwide. Every year, transportation authorities invest significant resources in improving road safety through better infrastructure, stricter traffic regulations, public awareness campaigns, and enhanced law enforcement. However, making effective decisions requires a clear understanding of accident patterns, contributing factors, and high-risk locations.

The objective of this project is to analyze historical road accident data to identify trends, risk factors, and accident-prone conditions using Microsoft Excel and Power BI. By transforming raw accident records into meaningful business insights, the project aims to support data-driven decision-making for transportation authorities, traffic police, urban planners, and road safety organizations.

The dataset contains detailed information about road accidents, including accident severity, number of casualties, vehicle types involved, road type, road surface conditions, weather conditions, light conditions, speed limits, urban or rural classification, and geographical locations. Analyzing these variables together helps uncover patterns that may not be immediately visible through raw data alone.

Rather than simply reporting accident counts, this project focuses on understanding why accidents occur, under what conditions they are most severe, and which factors contribute most to road safety risks. The insights generated from this analysis can help authorities prioritize safety initiatives, allocate resources more efficiently, and implement preventive measures in high-risk areas.

The project follows a complete Business Intelligence workflow beginning with data understanding, data cleaning, and transformation in Excel, followed by statistical analysis and interactive dashboard development in Power BI. Key Performance Indicators (KPIs), charts, trend analyses, and comparative visualizations will be used to present the findings in an executive-friendly format.

Business Problem

Despite the availability of large volumes of accident data, transportation authorities often face challenges in identifying the primary causes of accidents and determining where road safety improvements should be focused. Without proper analysis, decision-makers may struggle to prioritize investments, monitor accident trends, or evaluate the effectiveness of safety measures.

This project addresses these challenges by providing a centralized analytical view of accident data. It aims to identify accident hotspots, analyze the impact of road, weather, lighting, and speed conditions on accident severity, and highlight patterns that can support evidence-based policy decisions.

Project Objectives

The primary objectives of this project are:

- To analyze road accident trends across different time periods.
- To identify the major factors contributing to road accidents and casualties.
- To evaluate accident severity under different road, weather, lighting, and traffic conditions.
- To compare accident patterns across urban and rural areas.
- To identify high-risk locations requiring safety interventions.
- To develop interactive dashboards that enable stakeholders to explore accident data effectively.
- To provide actionable recommendations that can help improve road safety and reduce accident severity.

Stakeholders

The insights generated from this project can benefit several stakeholders, including:

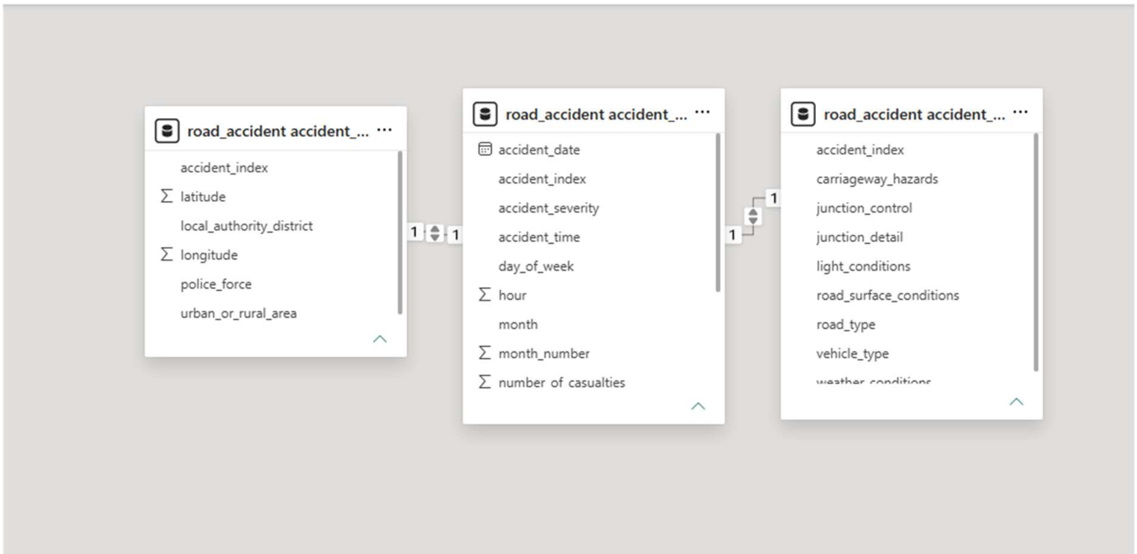
- Government Transportation Departments
- Traffic Police Authorities
- Road Safety Organizations
- Urban Planning Departments
- Highway and Infrastructure Authorities
- Policy Makers
- Emergency Response Agencies

These stakeholders can utilize the findings to improve road infrastructure, enhance traffic management strategies, optimize resource allocation, and implement targeted safety initiatives.

Expected Business Outcome

The expected outcome of this project is an interactive Business Intelligence dashboard that enables decision-makers to monitor accident trends, identify accident-prone conditions, evaluate safety performance, and make informed decisions based on data. The recommendations derived from this analysis can contribute to reducing accident frequency, minimizing casualties, and improving overall road safety.

- **Dataset Relationship Diagram**



- **List of Business Questions**

1. Which accident severity level occurs most frequently: Slight, Serious, or Fatal?
2. Which road type records the highest number of accidents?
3. Which road type is associated with the highest number of casualties?
4. Do weather conditions affect accident frequency and serious/fatal accident rate?
5. Are accidents more common during certain months or years?
6. Which light conditions are associated with higher accident counts?
7. Which vehicle types are linked with the highest number of casualties?
8. Which local authority/district has the highest number of accidents and casualties?

PHASE 2: DATA CLEANING AND PREPARATION (IN EXCEL)

- **Cleaned Excel file(s)**

https://drive.google.com/file/d/18pMges3n8apFG7FNrTf_AQrhj5IIHBqH/view?usp=sharing

- **Short Data Quality Notes (issues found + what you did about them)**

During the data preparation process, several data quality issues were identified and addressed to ensure accurate analysis:

- **Missing Values:** Some columns such as latitude, longitude, and weather conditions contained missing or blank values. These were either removed (if minimal) or handled using appropriate defaults or exclusions during analysis.
- **Inconsistent Text Formatting:** Columns like accident severity, road type, and weather conditions had inconsistent capitalization and spelling. These were standardized (e.g., converting all values to proper case) to ensure consistency in grouping and filtering.
- **Duplicate Records:** Duplicate accident records were checked using the accident_index field. Any duplicates found were removed to maintain data integrity.
- **Incorrect Data Types:** Some columns were initially stored as text (e.g., dates and times). These were converted into proper DATE and TIME formats to enable accurate time-based analysis.
- **Outliers and Invalid Values:** Certain numeric fields such as number of casualties and speed limit were reviewed for unrealistic values. Invalid entries were corrected or excluded where necessary.
- **Derived Columns Creation:** Additional columns such as year, month, hour, severity score, and serious/fatal flag were created to support better analysis and visualization.

These steps ensured that the dataset was clean, consistent, and ready for reliable analysis in Excel, SQL, and Power BI.

- **Data Dictionary (simple table)**

Data Dictionary Table

Column Name	Suggested Data Type	Source Table / Category	Description	Use in Analysis
accident_index	VARCHAR / Text	Main Identifier	Unique ID for each	Used as the primary key to identify each

			accident record	accident and connect tables
accident_date	DATE	accident_fact	Date on which the accident occurred	Used for monthly, yearly, and trend analysis
accident_time	TIME	accident_fact	Time when the accident occurred	Used for time-based accident analysis
day_of_week	Text	accident_fact	Day of the week when the accident happened	Used to analyze accident patterns by weekday
accident_severity	Text	accident_fact	Severity level of accident such as Slight, Serious, or Fatal	Used to analyze accident seriousness and risk
number_of_casualties	Integer	accident_fact	Number of casualties involved in the accident	Used to calculate total casualties and average casualties
number_of_vehicles	Integer	accident_fact	Number of vehicles involved in the accident	Used to understand vehicle involvement in accidents
speed_limit	Integer	accident_fact	Speed limit of the road where the accident occurred	Used to analyze accident patterns by speed limit

year	Integer	Derived Column / accident_fact	Year extracted from accident date	Used for yearly filtering and trend comparison
month	Text	Derived Column / accident_fact	Month name extracted from accident date	Used in monthly accident trend charts
month_number	Integer	Derived Column / accident_fact	Numeric month value from 1 to 12	Used to sort month names correctly in Power BI
hour	Integer	Derived Column / accident_fact	Hour extracted from accident time	Used for time-of-day analysis
severity_score	Integer	Derived Column / accident_fact	Numeric value assigned to severity, such as Slight = 1, Serious = 2, Fatal = 3	Used to support severity-based analysis
serious_fatal_flag	Integer	Derived Column / accident_fact	Flag value where 1 means Serious/Fatal and 0 means Slight	Used to calculate Serious/Fatal Accident Rate
latitude	Decimal	accident_location	Latitude coordinate	Used for geographical or

			of accident location	map-based analysis
longitude	Decimal	accident_location	Longitude coordinate of accident location	Used for geographical or map-based analysis
local_authority_district	Text	accident_location	District or local authority where the accident occurred	Used to identify high-accident districts
police_force	Text	accident_location	Police force area responsible for the accident record	Used for regional/police-force-based accident analysis
urban_or_rural_area	Text	accident_location	Indicates whether the accident happened in an urban or rural area	Used to compare urban and rural accident patterns
junction_control	Text	accident_conditions	Type of control at the junction, such as signals or give way signs	Used to study junction-related accident conditions
junction_detail	Text	accident_conditions	Details of the junction where the accident occurred	Used to analyze accident patterns by junction type

light_conditions	Text	accident_conditions	Lighting condition during the accident, such as daylight or darkness	Used to analyze accidents by visibility condition
carriageway_hazards	Text	accident_conditions	Hazards present on the carriageway, if any	Used to understand road hazard impact
road_surface_conditions	Text	accident_conditions	Surface condition of the road, such as dry, wet, snow, or frost	Used to analyze road condition risk
road_type	Text	accident_conditions	Type of road, such as single carriageway, dual carriageway, or roundabout	Used to identify accident-prone road types
weather_conditions	Text	accident_conditions	Weather condition at the time of the accident	Used to analyze weather impact on accident frequency and severity
vehicle_type	Text	accident_conditions	Type of vehicle involved in the accident	Used to analyze casualty

patterns by
vehicle type

PHASE 3: EXCEL ANALYSIS AND APPLICATION

- **Excel Analysis Workbook (.xlsx) with PivotTables, PivotCharts, and a Summary tab**

https://docs.google.com/spreadsheets/d/1dqqy7sY5pSkhJZARxWmJjC_LunZMhcMN/edit?usp=sharing&ouid=111051720778595000697&rtpof=true&sd=true

PHASE 4: SIMPLE STATISTICS (EXCEL ONLY-NO CODING)

- **Simple Statistics Summary**

The simple statistics analysis was performed in Excel using basic formulas such as `AVERAGE`, `MEDIAN`, `MIN`, `MAX`, `STDEV.S`, and `CORREL`. The purpose of this section is to understand the numerical pattern of the Road Accident dataset and explain what the results mean from a road safety point of view.

Three key metrics were selected for the analysis: **number of casualties**, **number of vehicles**, and **speed limit**. These metrics were chosen because they help explain the human impact of accidents, the number of vehicles involved, and the type of road environment where accidents occurred.

The **number of casualties** metric shows how many people were affected in each accident. The average and median values help identify the usual casualty level per accident, while the minimum and maximum values show the range of casualty impact. Most accidents involve a low number of casualties, but accidents with unusually high casualty counts are important because they may represent major accident cases requiring special attention.

The **number of vehicles** metric shows how many vehicles were involved in each accident. This helps understand whether accidents usually involve one or two vehicles or whether multi-vehicle accidents are common. Multi-vehicle accidents may be more complex and may require better traffic control, road safety awareness, and emergency response planning.

The **speed limit** metric was used to understand the road speed environment where accidents occurred. Speed limit is important because it may influence accident severity, but it may not explain casualty count alone. Other factors such as road type, vehicle type, weather conditions, light conditions, accident severity, and urban/rural location may also influence accident outcomes.

A correlation analysis was performed between **speed limit** and **number of casualties** using Excel's CORREL() function. This was done to check whether higher speed limits are associated with higher casualty counts. If the correlation value is close to zero, it means there is little or no strong linear relationship between speed limit and casualties. Therefore, road safety decisions should not be based only on speed limit; multiple factors should be analyzed together.

A monthly accident trend chart was also created to identify whether accidents are evenly distributed across months or whether some months show higher accident frequency. This helps in planning road safety campaigns, awareness programs, and enforcement activities during riskier periods.

Outliers were reviewed using the **number of casualties** column. High-casualty accidents were not automatically removed because they may represent real serious accident cases. These records are important for identifying high-risk situations and improving road safety planning.

Overall, the simple statistics analysis shows that most accidents have limited casualties and vehicle involvement, but some accidents have a much higher impact. The findings support the main objective of the project: to identify road accident patterns and help decision-makers reduce accident severity and casualty risk.

PHASE 5: BASIC SQL – UNDERSTANDING THE TABLES

- **SQL Script File (.sql) with all queries**

  https://drive.google.com/file/d/1GWspCyw49u7BIeTWweF6mKSNPggMbzt8/view?usp=sharing

- **Simple Schema Note:**

The Road Accident dataset was divided into three structured SQL tables for analysis. The main purpose of this schema is to separate accident-level facts, location details, and accident condition details while keeping a common key column to connect all tables.

Table Name	Purpose	Key Columns
accident_fact	Main table containing accident-level details, numerical values, severity	accident_index **(Primary Key)**, accident_date, accident_time, day_of_week, accident_severity, number_of_casualties,

	information, and time-related fields.	number_of_vehicles, speed_limit, year, month, month_number, hour, severity_score, serious_fatal_flag
accident_location	Stores geographical and location-related information for each accident.	accident_index **(Primary Key / Foreign Key)**, latitude, longitude, local_authority_district, police_force, urban_or_rural_area
accident_conditions	Stores road, weather, light, junction, surface, hazard, and vehicle-related condition details.	accident_index **(Primary Key / Foreign Key)**, junction_control, junction_detail, light_conditions, carriageway_hazards, road_surface_conditions, road_type, weather_conditions, vehicle_type

Relationship Note

- The common key column used to connect the three tables is:
- accident_index
- This column uniquely identifies each accident record.
- Main relationships:
- accident_fact.accident_index = accident_location.accident_index
- accident_fact.accident_index = accident_conditions.accident_index
- Here, accident_fact works as the main table, while accident_location and accident_conditions provide additional details about each accident.

Summary

- This schema supports structured accident analysis by separating the dataset into fact, location, and condition-based tables. It helps perform SQL joins, Excel analysis, and Power BI dashboard reporting more clearly. Using accident_index as the common key allows accident severity, casualties, road type, weather conditions, vehicle type, and location details to be analyzed together.

PHASE 6: POWER BI DASHBOARD DEVELOPMENT

- **Power BI Dashboard (.pbix)**

https://drive.google.com/file/d/1yiAMSkDyy-HSqa8pVN7lY2Yv86xe2-Ed/view?usp=sharing

- ## <u>Dashboard Screenshot</u>

- ## <u>Short Dashboard Notes</u>

The Power BI dashboard was created to analyze road accident patterns using the cleaned Road Accident dataset. The main objective of the dashboard is to identify accident severity, casualty impact, road-risk factors, and accident trends in a simple visual format.

The dashboard includes key performance indicators such as **Total Accidents**, **Total Casualties**, **Fatal Accidents**, **Serious/Fatal Accident Rate**, and **Average Casualties per Accident**. These KPIs provide a quick summary of the overall accident situation and help understand the scale and seriousness of road accidents.

The visualizations show accident patterns by **month**, **accident severity**, **road type**, **urban/rural area**, **vehicle type**, and **light conditions**. The monthly trend chart helps identify changes in accident frequency over time, while the severity chart shows how accidents are distributed across Slight, Serious, and Fatal categories.

The road type and vehicle type visuals help identify which road environments and vehicle categories are linked with higher accident counts or casualty levels. The urban/rural comparison helps understand whether accidents are more frequent in urban or rural areas. The light condition visual helps analyze whether accidents are more common during daylight or darker conditions.

Interactive slicers such as **Year**, **Accident Severity**, and **Weather Conditions** were added to make the dashboard more user-friendly. These filters allow users to focus on specific time periods, severity levels, or weather situations.

Overall, the dashboard supports road safety decision-making by highlighting accident trends, high-risk conditions, and casualty patterns. It can help authorities plan safety campaigns, improve traffic control, and focus on areas or conditions where accident risk is higher.